Equipment Market Signals: Do Input Costs and Rates Lead Equipment Prices?

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Question

Which cost and demand signals move with construction equipment prices, and with what delay? This demonstrates my approach using public data only. It does not reflect any company's actual pricing.

Data and method

- **Sources (BLS and Census via FRED, monthly, Jan 1993 to Aug 2026, 392 months):** PPI for construction machinery manufacturing (price proxy), PPI for steel mill products (input cost), total construction spending (demand proxy), and the federal funds effective rate (financing cost).
- **Pipeline:** Loaded raw files into Snowflake, joined them into one monthly table with SQL, and converted each series to year-over-year change so shared long-term trends would not create false correlations.
- **Analysis:** Used window functions to lag each signal from 0 to 24 months and `CORR()` to compare it with machinery price changes, across all months and since 2015 (about 140 months).
- **Data check:** My first steel series had missing months, cutting its usable history by more than half. I caught it because its month count was far below the other series, and replaced it with a complete series.

Findings

1. **Steel price changes lead machinery price changes by roughly a year.** The correlation is about 0.16 in the same month, 0.50 at 6 months, and peaks near 0.62 at 12 to 14 months, nearly identical in both time windows. Excluding 2020 to 2024, the 12-month correlation is 0.45 (versus 0.62 overall), so the relationship holds in normal conditions but is strongest during large cost shocks like 2021 to 2022.
2. **Interest rate changes move with machinery prices at the same time, not ahead of them.** The correlation is 0.39 in the same month (0.60 since 2015) and fades to 0.26 at 6 months. Both series rose together during the recent inflation cycle, so I read this as a shared cycle, not rates driving prices.
3. **Construction spending is a weak signal.** Correlations are 0.17 to 0.23 across all months and 0.25 to 0.31 since 2015.

Charts (provided separately): machinery vs. steel year-over-year change, standardized, with steel shown 12 months earlier; and correlation by steel lead time, peaking around 13 months.

Limits

Correlation is not causation. National PPI is a proxy for pricing, not company data. Year-over-year windows in neighboring months overlap, so the correlations look more certain than they are. Construction spending is in nominal dollars.

Next steps with internal data

Test product-line and regional pricing instead of one national index, add dealer and competitor data, adjust spending for inflation, and move from correlations to a model with controls. Refresh the Snowflake tables monthly so the analysis updates automatically.